PondLedger — Product Spec (v1)

A feed-and-profit tracker for catfish farmers Prepared by SitesbyFritz · 2026-08-07

The problem, in one line

For a catfish farmer, feed is 60–70% of total cost, and the two things that most decide whether a cycle makes money — feed conversion ratio (FCR) and real cost-per-kg — are invisible when records live in a notebook or in your head. By the time the fish are harvested, it's too late to act on them.

PondLedger makes those numbers visible *while the cycle is still running*, so decisions about feeding and harvest timing can be made with facts instead of guesswork.

Who it's for

A single catfish farmer running one or several ponds/tanks at different stages, on a phone, often with patchy internet. Not a lab. Not a corporate hatchery. The design assumption is: if it takes more than 30 seconds a day, it won't get used.

The core loop

Each day, per pond, the farmer logs three things:

1. **Feed given** — kilograms and cost (₦)
2. **Mortalities** — number of fish lost
3. **Sample weight** — average weight of a few fish, entered periodically (not daily)

From just those inputs, PondLedger computes what the farmer currently cannot see:

- **Live FCR** — feed used ÷ weight gained, updated as the cycle runs
- **Cost-per-kg in naira** — the true break-even floor for this pond
- **Break-even vs. market price** — know your floor *before* a middleman names a number at the farm gate
- **Harvest-timing signal** — flags the point where extra feed stops paying for itself
- **Cycle profit/loss report** — compare pond-to-pond and cycle-to-cycle

Why this idea (and not a full "farm management" app)

Research across Nigerian catfish studies is consistent: the highest-impact profit levers are selling price, FCR, and feed price — in that order. Water-quality sensors, marketplaces, and IoT feeders are real but are either expensive, network-dependent, or solve problems a single farmer can't move alone. Feed efficiency is the one big lever the farmer controls every single day. Build there first.

A quiet second benefit: a farmer who knows his cost-per-kg has real negotiating power against middlemen who currently dictate farm-gate prices.

Worked example (illustrative — real numbers get plugged in)

> One pond, 1,000 fingerlings, 10% mortality, average harvest weight 1.0 kg → 900 kg biomass. FCR 1.5, feed at ₦1,300/kg.

Item	Value
Feed used	1,350 kg
Feed cost	₦1,755,000
Total cost (feed ≈ 65%)	₦2,700,000
Cost-per-kg (break-even)	**₦3,000/kg**
Middleman offer @ ₦2,900/kg	Revenue ₦2,610,000 → **loss ₦90,000**
Direct sale @ ₦3,600/kg	Revenue ₦3,240,000 → **profit ₦540,000**

The point isn't these exact figures — it's that the same pond swings from loss to profit on price and FCR, and PondLedger surfaces that *before* the sale, not after.

MVP scope (v1) — deliberately narrow

In:

- One farmer, one phone, mobile-first
- Add/manage multiple ponds, each with its own cycle (stocking date, count, fingerling cost)
- Daily quick-entry: feed (kg + ₦), mortalities, optional note
- Periodic sample-weight entry
- Auto-calculated dashboard per pond: FCR, cost-per-kg, survival %, days in cycle
- Break-even vs. a manually-entered current market price
- End-of-cycle P/L summary
- Works offline; syncs when back online

Out (v2 and later):

- Water-quality sensors / IoT feeders
- Multi-user / staff accounts
- Marketplace or buyer directory
- Disease diagnosis
- SMS/WhatsApp alerts

Data model (starting point)

- **Farm** — owner, currency (₦), default feed price
- **Pond** — name, type (tank/earthen/plastic), status
- **Cycle** — pond, stocking date, fish count, fingerling cost, expected harvest window

- **FeedLog** — cycle, date, feed kg, feed cost
- **MortalityLog** — cycle, date, count
- **WeightSample** — cycle, date, average weight
- **Harvest** — cycle, date, total kg, revenue, buyer type (middleman/direct/retail)

Derived (not stored): FCR, cost-per-kg, survival %, projected profit.

Suggested build stack

- **Front end:** Framer or a Claude Code React app — mobile-first, big tap targets, minimal typing
- **Backend/DB:** Supabase (Postgres + auth); offline-tolerant with local cache and sync
- **Charts:** lightweight (cost-per-kg over time, FCR trend)

Start as a private tool for one farm. If it earns its keep, the same template becomes a product for the large, mostly-paper catfish market — a natural fit for the SitesbyFritz ICP.

Success test for v1

One question decides if v1 worked: *at any moment, can the farmer open the app and say what a kilo of fish in each pond is costing him right now?* If yes, everything else is upside.

Next step options: a clickable prototype of the daily-entry screen + dashboard, or a v1 build plan with the Supabase schema and screens.